FIM 590
Data Analytics with SQL
Fall 2026
Monday 6:00 PM
1 Credit Hour

Instructor Name: Robert Graham Carroll Jr
E-mail: rgcarrol@ncsu.edu

Course Description

This course explores how we can utilize various technological tools to approach data collection and analysis. Students will be required to analyze various types of financial data, with an emphasis on programming best practices and on validating query results.

The course will set the foundation for participants to be able to develop SQL code in various programming environments. Upon finishing the course, participants will have a body of SQL work, kept in their own GitHub repository, that they can describe on their resumes.

Weekly Schedule

- Week 1: Introduction to course, Sample Data Set, SQL Basics (SELECT, WHERE, ORDER BY, LIMIT); Assignment 1: Equity Price Exploration
- Week 2: Aggregations and GROUP BY, WRDS intro (class demo only), stock returns and trading volume; Assignment 2
- Week 3: Data Preparation, JOINs, equity prices and company fundamentals; Assignment 3
- Week 4: CTEs and CASE, FX Data Set, introduction to GitHub Copilot; Assignment 4
- Week 5: Dates and financial calculations, Fixed Income Data Set, Python interaction with SQL; Assignment 5
- Week 6: Window functions, rolling returns and volatility, SQL engine performance; Assignment 6
- Week 7: Final in class assignment (integrated analysis on a multi-asset data set) and written test -or- project presentation (will poll students first week to see what they prefer)

Topics and their order are tentative and may be adjusted as the course progresses. Course SQL is written for DuckDB, so vendor-specific topics such as SQL Server stored procedures are covered only as class demonstrations.

Grading

The course will consist of both individual programming assignments and project based individual work.

- Class Assignments: 70%
- Final Assignment: 30%

Assignments are graded on the version last pushed to your repository before the deadline. Grading is based on query results (columns, rows and, where required, row order), with partial credit, not on SQL style. Work that validates its results is rewarded.

Late policy: None.

Tools and Submission

Each student works in a private GitHub repository generated from the course template. Assignments are submitted by pushing to that repository. The automated feedback you see in GitHub Actions and the local practice check is for practice only; the official grade is computed by the instructor from the submitted query files.

You will need a GitHub account, Visual Studio Code and Python 3. Setup instructions are in the course materials repository. All assignments can be completed offline with the provided practice data, which is synthetic. WRDS is used for class demonstrations only.

AI and GitHub Copilot Policy

- Assignments 1-3: you write all SQL yourself; Copilot must be turned off.
- Assignment 4: Copilot is introduced in class, with clear boundaries.
- Assignment 5: Copilot is optional. Assignments 6-7: Copilot is allowed.
- Whenever AI is allowed, you are responsible for understanding, validating and being able to explain every query you submit. No grade depends on having Copilot access.

Academic Integrity and University Policies

All work must be your own except where AI assistance is explicitly allowed above. Students are bound by the NC State Code of Student Conduct (POL11.35.01). Accommodations for students with disabilities are handled through the Disability Resource Office (REG02.20.01).